

Mock test 4.

1. x va y natural sonlar uchun $x + y = 19$ bo'lsa, xy ning eng katta qiymatini toping.

- A) 80
- B) 84
- C) 90
- D) 100

2. $n^2 + 2n$ soni juft son bo'lsa, quyidagi ifodalardan qaysi birining qiymati albatta juft bo'ladi?

- A) $n + 1$
- B) $3^n + n$
- C) $n^2 + n + 1$
- D) $n^2 + 4$

3. a va b raqamlar uchun $E_k(\overline{aa}; \overline{bb}) = 132$ bo'lsa, $a + b$ ning eng katta qiymatini toping.

- A) 7
- B) 8
- C) 9
- D) 10

4. $1^2 + 2^2 + 3^2 + \dots + 99^2$ ifodaning oxirgi raqamini toping.

- A) 0
- B) 4
- C) 6
- D) 9

5. Sotuvchi do'konidagi mahsulotning 80% ini 20% foyda bilan, qolganini 10% foyda bilan sotdi. Bunga ko'ra sotuvchi necha % foyda qildi?

- A) 14
- B) 16
- C) 18
- D) 19

6. Oralaridagi masofa 180 km bo'lgan ikki shaharlardan biridan ikkinchisiga qarab avtobus, oradan 30 minut o'tgach esa ikkinchisidan birinchisiga qarab yengil avtomobil yo'lga chiqdi. Agar avtomobilning tezligi avtobusning tezligidan 15 km/soatga ko'p, hamda ular yo'lining o'rtasida uchrashishgan bo'lsa, avtobus tezligi necha km/soat?

- A) 40
- B) 45
- C) 50
- D) 60

7. 600 kg yangi uzilgan meva tarkibida 85% suv bo'ladi. Ma'lum vaqtdan keyin meva tarkibida 80% suv bo'lgan bo'lsa, endi mevaning og'irligi qancha kg?

- A) 300
- B) 350
- C) 400
- D) 450

8. Koordinatalar tekisligida $P(-2\sqrt{3}; 6)$ nuqta va koordina boshini tutashtiruvchi chiziq OP ni o'zgarmas bo'lish sharti bilan soat strelkasi yo'nalishi bo'yicha 150° ga aylantirilganda hosil bo'ladigan koordinata qanday bo'ladi?

- A) $(-2\sqrt{3}; -6)$
- B) $(2\sqrt{3}; 6)$

C) $(2\sqrt{3}; -6)$

D) $(6; -2\sqrt{3})$

9. $\log_{\sqrt{2}} 8 + \log_9 27 + \log_{\frac{1}{5}} 25$ hisoblang.

A) 4

B) 4,5

C) 5

D) 5,5

10. O'rganilgan narsani unutish miqdorini aniqlash uchun matematik model yaratildi. Ushbu modelga ko'ra, birinchi o'tkazilgan testdan olingan ball A. Agar birinchi testga o'xshash test t oydan so'ng qo'llanilsa, ushbu testdan olish mumkin bo'lgan ball $S(t)$ sifatida baholanishi mumkin. S ning vaqtga bog'liq qiymati quyida keltirilgan:

$$S(t) = A - 15 \lg(t+1) \text{ va } \frac{A - S(t)}{A} \text{ ifoda esa unutish ulushiga mos keladi.}$$

Bunga ko'ra birinchi sinovdan 75 ball olgan Behruz 9 oydan so'ng necha foiz bilimini unutadi?

A) 16

B) 18

C) 20

D) 24

11. $\log_2(5x-13) < 5$ tengsizlikni qanoatlantiruvchi eng kichik va eng katta butun sonlarning yig'indisi qanday?

A) 10

B) 11

C) 12

D) 13

12. Ayirmasi d, dastlabki n ta hadi yig'indisi S_n bo'lgan arifmetik progressiya uchun $\frac{S_8 - S_6}{S_9 - S_5}$ ifodaning qiymatini qanday?

A) 2

B) $\frac{3}{2}$

C) 1

D) $\frac{1}{2}$

13.
$$\begin{cases} x^2 + 1 = 2y \\ y^2 + 1 = 2z \\ z^2 + 1 = 2x \end{cases}$$
 tenglamalar sistemasidan $x + 2y + 3z$ ning qiymatini toping.

A) 0

B) 3

C) 6

D) 12

14. 101225 sonning raqamlarini almashtirish natijasida hosil qilingan olti xonali natural sonlarning nechitasi 5 ga qoldiqsiz bo'linadi?

A) 48

B) 50

C) 54

D) 56

15. $y = \sqrt{x^2 - 9x + 18}$ funksiyaning aniqlanish sohasiga tegishli nechta manfiy butun son $(x^2 - 25)(x^2 + 4x) \leq 0$ tengsizlikni qanoatlantiradi?

- A) 1
B) 2
C) 3
D) 4

16. $\frac{x^2 + 5x + 6}{x + 5} + \frac{x^2 + 3x - 4}{x + 3} \geq 2x$ tengsizlikning manfiy butun yechimlari nechta?

- A) $\emptyset$
B) 1
C) 2
D) 3

17. $f(x) = \begin{cases} x^2 + m, & x < 1 \\ 5, & x = 1 \\ mx + n, & x > 1 \end{cases}$ funksiya haqiqiy sonlar to'plamida uzluksiz aniqlangan bo'lsa, mn ning qiymati nechaga teng?

- A) 4
B) 3
C) 1
D) -1

18. Ikkinchi darajali $P(x)$ ko'phad uchun quyida berilganlardan foydalanib, $P(1) + P'(-3)$ yig'indining qiymatini hisoblang

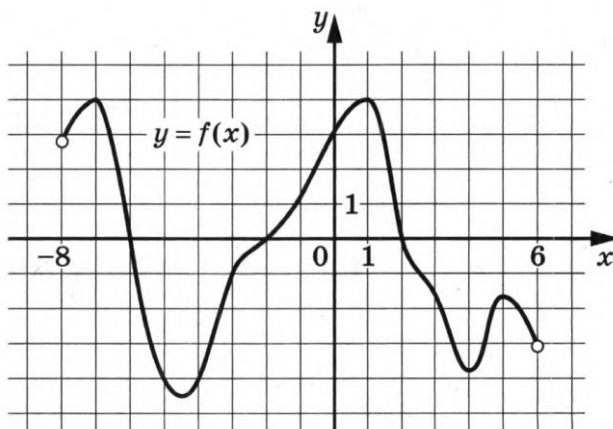
I. x ning har bir qiymatida $P(x) \geq 0$

II. $P(-2) = 0$

III. $P'(-1) = 4$

- A) 10
B) 12
C) 14
D) 15

19. Rasmda $(-8; 6)$ oraliqda aniqlangan $y = f(x)$ funksiyaning grafigi ko'rsatilgan. $f(x)$ funksiyaning hosilasi manfiy bo'lgan butun nuqtalar sonini aniqlang.



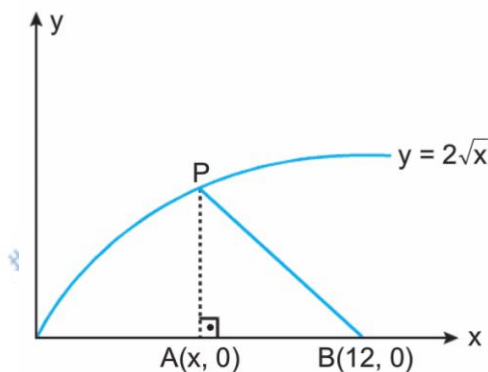
- A) 4
B) 3
C) 5
D) 2

20. $f(x) = 1 + x + x^2 + x^3 + \dots + x^n$ funksiya uchun $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = 120$ o'rinli bo'lsa, $f(1)$ nechaga teng?

$$\left(\text{eslatma : } f'(x) = \lim_{x \rightarrow a} \frac{f(x + \Delta x) - f(x)}{x - a}; \Delta x = x - a \right)$$

- A) 12
B) 13
C) 14
D) 16

21. Rasmda $y = 2\sqrt{x}$ egri chiziq va A(x; 0) va B(12; 0) nuqtalar berilgan. Shunga ko'ra x ning qanday qiymatida ABP to'g'ri burchakli uchburchakning yuzi eng katta bo'ladi?



- A) 2
B) 3
C) 4
D) 5

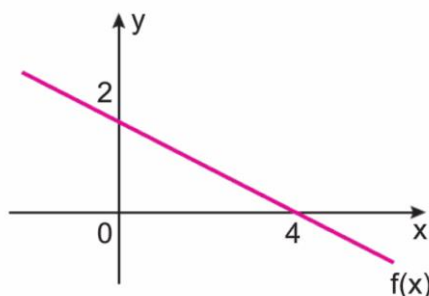
22. $|3|x| + x| = 20$ tenglama ildizlari yig'indisini toping.

- A) 5
B) -5
C) 0
D) 14

23. $\int_0^1 (\sqrt{1-x^2} + x - 1) dx$ ifodaning qiymati nechaga teng?

- A) $\frac{\pi - 2}{4}$
B) $\frac{\pi}{4}$
C) $\pi + 2$
D) $\frac{\pi + 2}{4}$

24. Koordinatalar tekisligida $f(x)$ chiziqli funksiya berilgan, bunga ko'ra $\int_0^2 x \cdot f(x) dx$ ifodaning qiymati qanday?



A) 3

B) $\frac{8}{3}$ C) $\frac{7}{4}$

D) 1

25. $A = \sin(15\pi - x) + \cos\left(\frac{17\pi}{2} + x\right)$; $B = \operatorname{tg}(x - 9\pi)$ berilganlarga ko'ra $A + B$ ning qiymati quyidagilardan qaysi biriga teng?

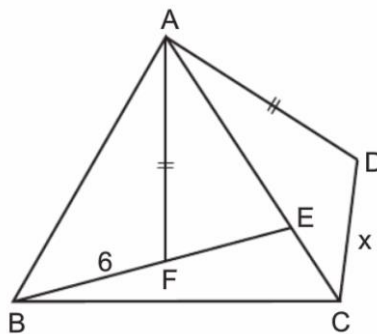
A) $\sin x$ B) $\cos x$ C) $\operatorname{tg} x$ D) $\operatorname{ctg} x$

26. $\operatorname{arctg}(x+1) + \operatorname{arctg}(x-1) = \operatorname{arctg} \frac{8}{31}$ tenglamani qanoatlantiruvchi musbat son quyidagi oraliqlardan qaysi birida joylashgan?

A) $\left[\frac{1}{4}; 1\right)$ B) $[1; 3)$ C) $[3; 5)$ D) $\left[5; \frac{9}{4}\right]$

27. ABC teng tomonli uchburchakda B, F E nuqtalar bir to'g'ri chiziqli yotadi va

$\angle BAC = \angle DAF$; $|AF| = |AD|$; $|BF| = 6$. Berilganlarga ko'ra $|CD| = x$ kesma necha sm ga teng?



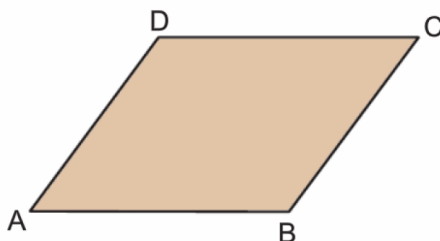
A) 2

B) 3

C) 4

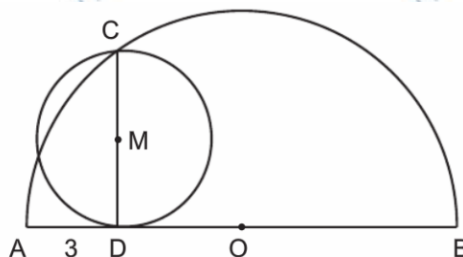
D) 6

28. ABCD parallelogramm shaklidagi quruqlikda A nuqtadan CD tomonning o'rta nuqtasiga va D nuqtadan BC tomonning o'rta nuqtasiga chiziqli to'siqlar qurish orqali to'rt xil maydon hosil qilindi. Agar eng katta maydon eng kichik maydondan $300 m^2$ ga ko'p bo'lsa, ABCD parallelogrammning yuzasi necha m^2 ga teng?



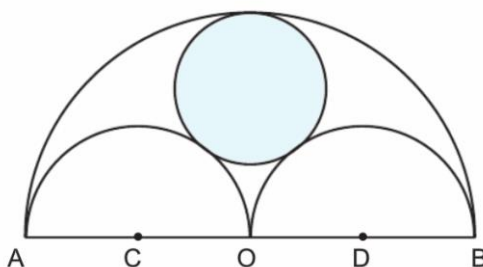
- A) 480
B) 520
C) 560
D) 600

29. Shaklda O markazli yarim aylana va M markazli butun aylana tasvirlangan. C, M, D bir to'g'ri chiziqda yotadi, $|AD| = 3\text{ sm}$, $|DC| = 6\text{ sm}$ berilganlarga ko'ra O markazli aylananing diametri necha sm ga teng?



- A) 8
B) 9
C) 10
D) 15

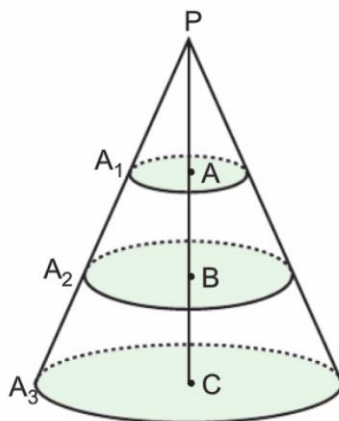
30. Shakldagi doiralar bir birga urinadi. C, O, D yarim doiralar markazlari. $AB = 24\text{ sm}$ bo'lsa, bo'yalgan doiraning yuzi necha sm^2 ga teng?



- A) 16π
B) 12π
C) 9π
D) 6π

31. Asosining yuzi A_3 bo'lgan quyidagi vertikal konusda yuzi A_2 va A_1 bo'lgan asosga parallel kesimlar o'tkazildi.

$|PC| = 3|AB| = 6|BC|$ berilgan bo'lsa, $\frac{A_1}{A_3}$ nisbat nimaga teng?



- A) $\frac{3}{4}$
B) $\frac{9}{16}$

C) $\frac{9}{25}$

D) $\frac{1}{4}$

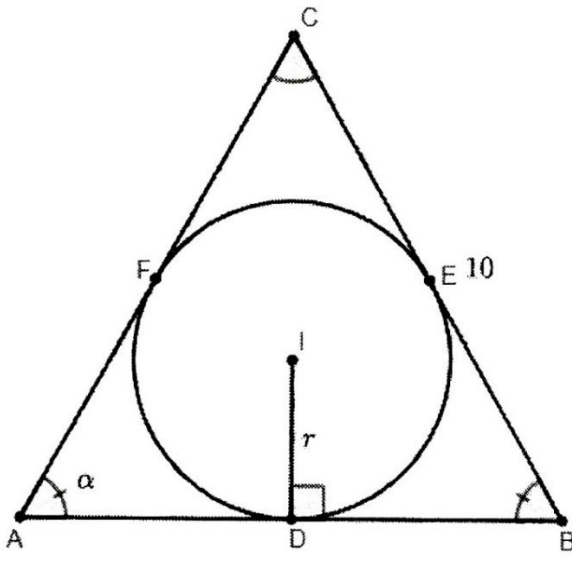
32. $A(4x+2, 2)$ va $B(10; x)$ nuqtalar berilgan. O nuqta koordinatalar boshi va $\overrightarrow{OA}$ va $\overrightarrow{OB}$ vektorlar kollinear bo'lsa, x ning qabul qiladigan qiymatlari yig'indisi topilsin.

A) -1

B) $-\frac{1}{2}$

C) 0

D) $\frac{1}{2}$

Topshiriq(33-35)lar va javob variant(A-F)larni o'zaro moslashtiring.	Javoblar
ABC teng yonli uchburchak bo'lib, $BC = 10$; $\sin \alpha = \frac{12}{13}$ ga teng.  33. ABC uchburchakka ichki chizilgan aylana radiusi nechaga teng? 34. ABC uchburchakka tashqi chizilgan aylana radiusi nechaga teng? 35. ABC uchburchakning yuzi nechaga teng?	A) $\frac{10}{3}$ B) $\frac{100}{39}$ C) $\frac{65}{12}$ D) $\frac{6000}{169}$ E) 17 F) 40

36. $m \in R$ va $2 \leq m \leq 6$ tekislik biror nuqtadan m uzoqlikdagi nuqtalar to'plami bilan qoplangan?

a) berilganlar bo'yicha hosil bo'lgan figuraning yuzini toping.

javob a) _____

b) biror nuqtadan m uzoqlikda hosil qilingan aylanalarning eng kichigining uzunligini toping.

javob b) _____

37. $2 \sin^2 \left(\frac{x}{2} - \frac{\pi}{4} \right) \cdot \sin^2 \left(\frac{x}{2} + \frac{\pi}{4} \right) = \cos^4 x$ tenglama berilgan.

a) Tenglamaning $[0; 2\pi]$ kesmadagi ildizlari soni nechta?

javob a) _____

b) Tenglamaning eng kichik musbat ildizi x_0 bo'lsa, $\frac{\pi}{x_0}$ ning qiymati qanday?

javob b) _____

38. $\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \dots + \sqrt{1 + \frac{1}{999^2} + \frac{1}{1000^2}}$ ifoda berilgan.

a) ifodaning qiymatini hisoblang.

javob a) _____

b) ifodaning qiymatini o'nli kasr ko'rinishida yozilganda nechta to'qqizdan iborat bo'ladi?

javob b) _____

39. $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ parametrli tenglama berilgan. Tenglamaning ikkala ildizi ham 3 dan katta.

a) a ning eng kichik butun qiymatini toping.

javob a) _____

b) a ning shu qiymatida tenglamaning ildizlari yig'indisini toping.

javob b) _____

40. $f(2x+1) - 3f(1-2x) = x^2 + 2x$ haqiqiy sonlar to'plamida aniqlangan funksiya berilgan.

a) $-8f(x) + 4x - 3$ ning ko'rinishi qanday?

javob a) _____

b) $f(3)$ ning qiymatini toping.

javob b) _____

41. $\sin x = \frac{x}{100}$ tenglama berilgan.

a) Tenglama nechta ildizga ega?

javob a) _____

b) Tenglamaning ildizlari yig'indisini toping.

javob b) _____

42. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{99}{100!}$ ifodaning yig'indisi S ga teng.

a) $S + \frac{1}{100!}$ ning qiymati nechaga teng?

javob a) _____

b) $\frac{1}{1-S}$ ifodaning qiymati nechta nol bilan tugaydi?

javob b) _____

43. $9x^2 - px - 4 = 0$ tenglamaning ildizlari $\sin x$ va $\cos x$ bo'lsin, $p > 0$.

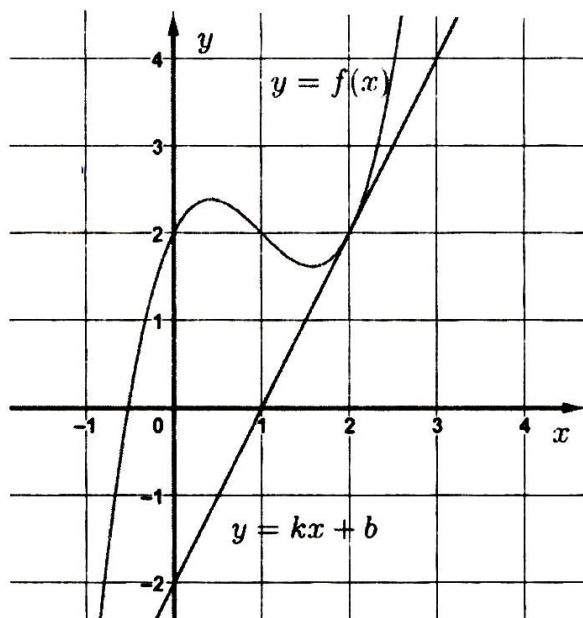
a) p ning qiymatini aniqlang.

javob a) _____

b) $9x^2 - px - 4$ kvadrat uchhadning koeffitsiyentlari yig'indisini toping.

javob b) _____

44. Grafikda $y = f(x)$ funksiya va uning $x_0 = 2$ nuqtasidan o'tkazilgan urinmasi tasvirlangan. Agar $g(x) = x^2 f(x)$ bo'lsa:



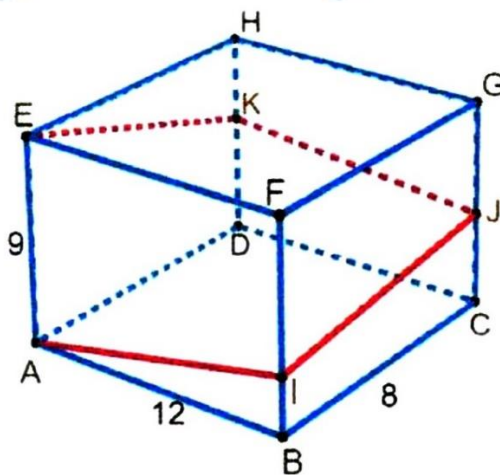
a) $f'(2)$ ning qiymatini toping.

javob a) _____

b) $g'(2)$ ning qiymatini toping.

javob b) _____

45. ABCDEFGH- to'g'ri burchakli parallelepiped; $AB=12$; $BC=8$; $AE=9$ bo'lsa:



a) $AI+IJ+JK+KE$ yig'indining eng kichik qiymati qanday?

javob a) _____

b) ABCDEFGH Jismning to'la sirtini toping.

javob b) _____